Groups represented by incidence geometries

Complete written comparison: two originals and one shared theory

The complete replacement is longer in the fixed primary measure:

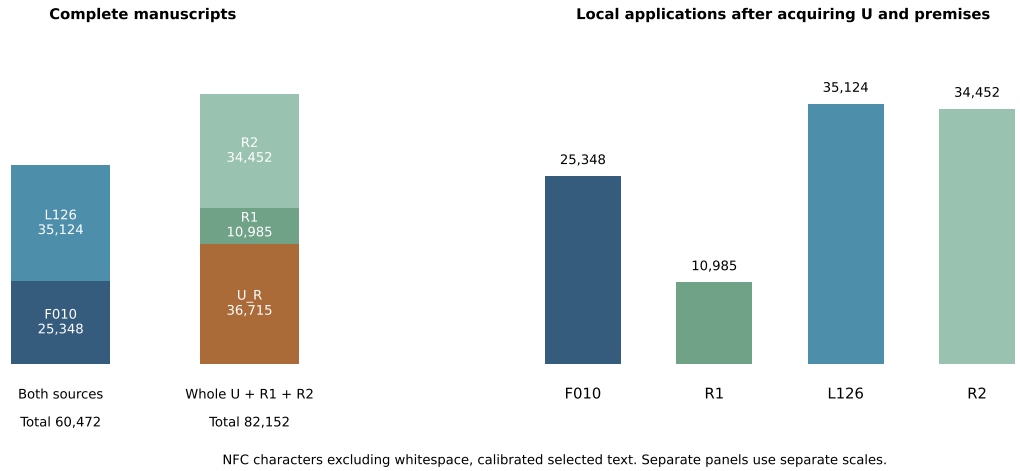
$$\frac{c(\mathrm{F010}) + c(\mathrm{L126})}{c(U) + c(R_1) + c(R_2)} = \frac{25\,348 + 35\,124}{36\,715 + 10\,985 + 34\,452} = \frac{60\,472}{82\,152} \approx \mathbf{0.736099}.$$

This is expansion, not achieved whole-topic compression. The whole common proof is paid once, including the finite-normal-pair existence proof now shared by both applications. Neither application is an uncharged prerequisite of its sibling.

The fixed primary coordinate is *calibrated selected text, NFC, characters excluding whitespace*. It preserves the prior F010 source policy. All pages and references are included: two sources of 13 pages each, common theory 13 pages, and applications of 4 and 12 pages. Exact running furniture is removed and bound glyph aliases are applied; raw, untrimmed text remains an explicit sensitivity.

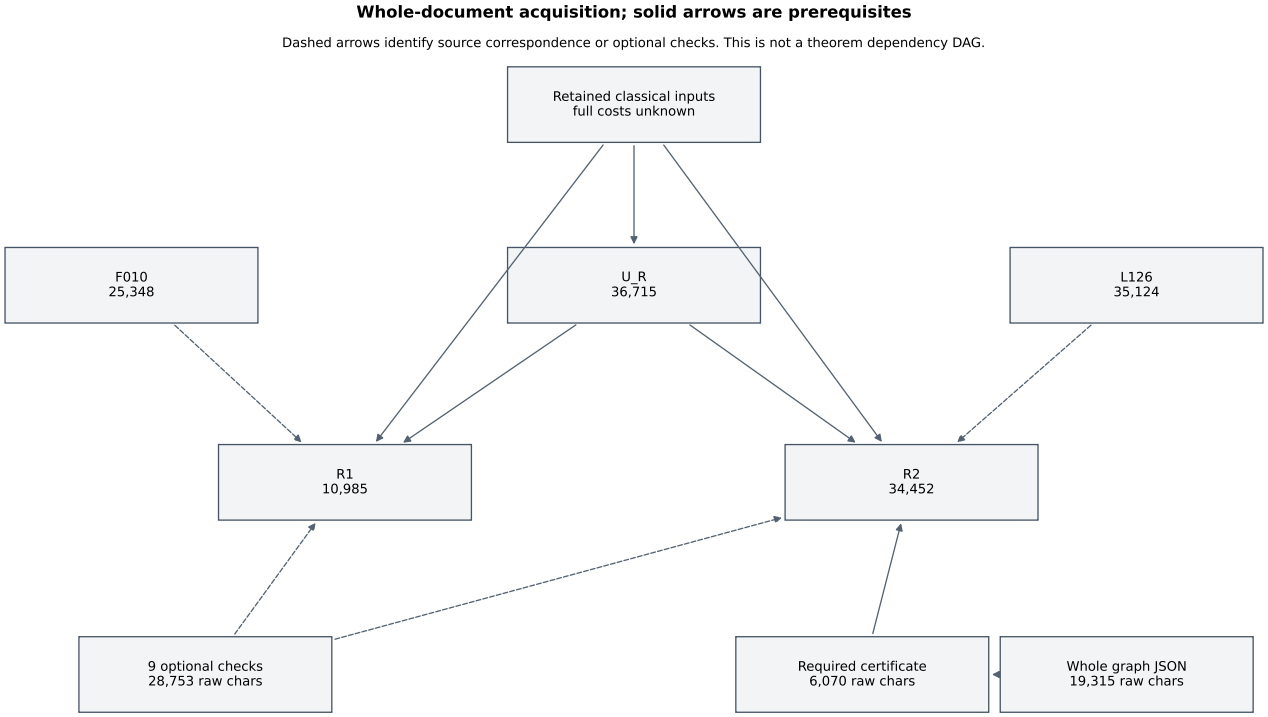
Conditional reuse	Source	Complete local paper	Factor
F010 after whole U and premises	25 348	10 985	2.307510
L126 after whole U and premises	35 124	34 452	1.019505

These conditional factors describe reuse *after* acquiring the whole common theory and actual retained classical premises. L126 also requires the complete graph certificate and graph JSON described on page 2. The nine optional checks are not acquisition prerequisites. Conditional reuse does not undo the growth of the complete shared proof.



Scope of the numbers. These are manuscript lengths, not minimal proof lengths, mathematical difficulty, or total research effort. Executable listings are excluded on both sides; written mathematical algorithms and proofs remain. Software-inclusive factors are unknown, not zero, because matched source software and all outside proof costs are unavailable.

What has to be acquired

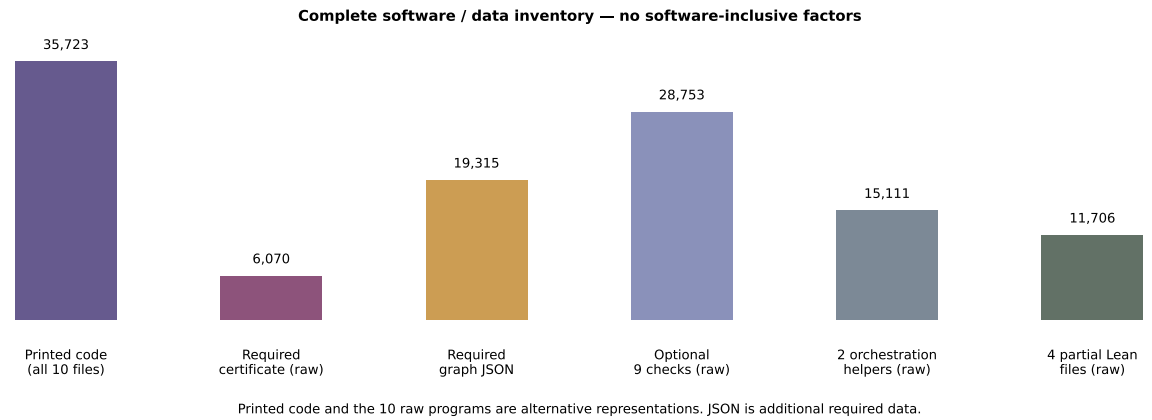


The diagram shows whole-document acquisition and evidence association, not an internal theorem dependency graph. Its box counts carry no compression factor. Starting with the complete common paper costs 36 715; adding all of R_1 reaches 47 700; adding all of R_2 reaches 82 152. All local context and bibliography are paid.

Required finite premise for L126. The complete `certificate/check.py` is 8 030 bytes (6 070 primary raw characters), and the full `certificate/graph.json` is 19 326 bytes (19 315 primary raw characters). The elementary certificate establishes the full group of the 896-vertex graph used in the Section 8 counterexample. Neither a summary nor the printed code appendix substitutes for the complete graph data. These are actual proof inputs even though manuscript ratios inventory their cost separately.

Optional and separate evidence. Nine complete validation programs contain 35 601 bytes (28 753 primary raw characters). The 16-page appendix prints all ten programs; its 35 723 primary characters and the ten raw programs' 34 823 characters are alternative representations, never added twice. The graph JSON remains separate. Two orchestration helpers total 16 622 bytes and four partial Lean files 15 931 bytes. Their 22 declarations do not formalize either complete source or the graph instance. Runtime, libraries, outside proofs and full graphical structure retain unknown costs.

Inventory, sensitivities and attribution



All 30 combinations of three text policies, NFC/NFKC and five units are reported in `results.json`, with exactly 30 whole-written and 60 conditional fractions. The selected calibrated NFC slice is:

Unit	Whole written	F010 conditional	L126 conditional
Whitespace tokens	0.768156	2.366350	1.106760
Alphanumeric units	0.809699	2.654554	1.108602
Nonspace characters	0.736099	2.307510	1.019505
Collapsed characters	0.741529	2.317469	1.034055
Collapsed UTF-8 bytes	0.711821	2.265009	1.003041

The complete old F010 source vector is unchanged. Every current replacement PDF, including the changed four-page R_1, was newly extracted in the accepted input packet. This numerical packet performs no new extraction, font reconstruction, scientific computation or proof review. No alternative unit is substituted for the fixed primary. The full raw, untrimmed NFC nonspace coordinate gives 0.749513 for the whole comparison; it is a sensitivity, not a replacement headline.

Exact source boundary. F010 is Díaz Ramos–Molinier–Viruel, *Every finite group is represented by a finite incidence geometry*, arXiv:2511.02410v2, 16 December 2025. L126 is Leemans–Stokes–Tranchida, *Groups represented by incidence geometries*, arXiv:2506.10464v1, June 2025. The accepted scientific contract retains all six F010 blocks and 31 L126 dispositions, including explicit finite-field and small-case corrections and the chamber-connectivity qualification for splitting. The broader Section 6.2 and Section 8 programmes are not claimed solved. Classical projective, Tits, Frucht and named-group premises remain attributed and unpriced.

Review boundary. The scientific successor, proof relocation and input boundary have distinct accepted reviews. This author numerical packet requires its own independent arithmetic and reader review. It claims no new source, question, formalization or historical-priority credit. Earlier frozen editions remain intact.